

Aditya Thapliyal

Dehradun, India

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Education

Graphic Era Hill University, Dehradun

Bachelor of Technology in Computer Science and Engineering

2023 – Present

CGPA: 8.7 / 10

Army Public School, Jodhpur

Class XII (CBSE), 2022

96.00%

Class X (CBSE), 2020

95.67%

Technical Skills

Languages: C++ (DSA, STL), JavaScript, SQL, Java.

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Theory of Computation, Object Oriented Programming (OOP).

Web & Backend: HTML, CSS, JavaScript, Node.js, Express.js

Databases: MongoDB, MySQL

Cloud: EC2, S3, VPC

ML & Data Science: Pandas, NumPy, Scikit-learn, Matplotlib

Developer Tools: Git, GitHub, VS Code, Postman, Unix, Linux.

Projects

Connect-V-2.0 (GitHub Repository)

Jan 2026 – Aug 2026

Tech Stack: React, Node.js, Express.js, FastAPI, MongoDB, ChromaDB, MinIO, SentenceTransformers

- Built a full-stack student collaboration platform using React, Node.js, Express, FastAPI, MongoDB, and ChromaDB for **AI-powered team formation** and project collaboration.
- Developed a semantic recommendation engine using **SentenceTransformers**, **ChromaDB**, **dense vector embeddings**, and **cosine similarity** to match users beyond exact keyword searches.
- Integrated **MinIO pre-signed URL uploads** for scalable media storage and developed a dedicated **FastAPI microservice** for vector indexing and semantic search.

RedisLite (GitHub Repository)

May 2026 – Aug 2026

Tech Stack: C++17, STL, std::unordered_map, std::priority_queue, std::chrono, File I/O

- Built a Redis-inspired in-memory key-value store in C++ supporting **CRUD operations**, **TTL-based key expiration**, **snapshot persistence**, and an interactive command-line REPL.
- Implemented a hybrid TTL expiration strategy combining **lazy deletion** with a **min-heap** scheduler while preserving expiration accuracy across restarts using **absolute epoch timestamps**.
- Designed a **snapshot manager** that serializes database state to disk and automatically restores it on startup using a custom serialization format with robust recovery from malformed snapshot records.

Esti-MATE (GitHub Repository)

Jul 2026 – Aug 2026

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Flask, Streamlit, HTML, CSS, JavaScript

- Built an end-to-end ML web application to predict Bengaluru house prices using **Linear Regression** on a Kaggle dataset (**13,000+ records**), achieving an **84.52% R²** score.
- Performed data preprocessing and feature engineering including missing value handling, BHK extraction, one-hot encoding, and statistical outlier removal using Pandas.
- Compared **Linear Regression**, **Lasso**, and **Decision Tree** using **GridSearchCV** with 5-fold cross-validation, selecting Linear Regression as the best-performing model.

Achievements & Activities

- Qualified **GATE 2026 (Computer Science)** AIR : 5820, SCORE : 529 (97.24 percentile) (Scorecard)
- Salesforce Developer and Automation with Agentblazer champion (Link)
- Top-15 Finalist (**100+ teams**) at Graphethon 2025 for an Alzheimer's memory-assistance solution.